Google I/O Extended

Mobile

Dilum De Silva

Mobile

The Folding Phone

New **Google Pixel Fold** combines the helpfulness of Pixel with a design that unfolds into a compact tablet, unlocking even more ways for you to use your device.

Mobile

What’s new in Mobile-Tech?

New in Android

New in Flutter & Dart

New in Firebase

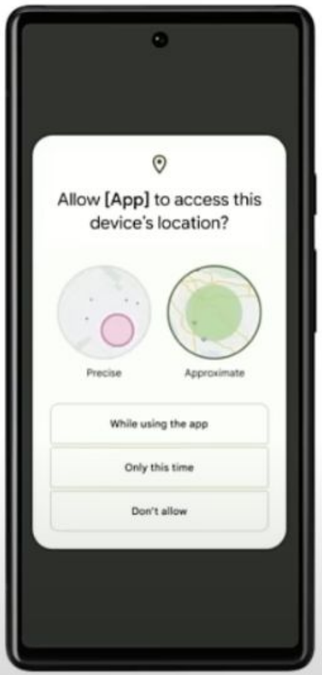
New in Material Design

New in Android

Improved privacy controls for users & also for devs

Indicators

Privacy Dashboard



Approximate Location

Privacy Nudges

Camera/Mic Toggles

New in Android

Android 14 is big on privacy

Partial media access

Adopt the Photo Picker or you can manually check for these permissions:

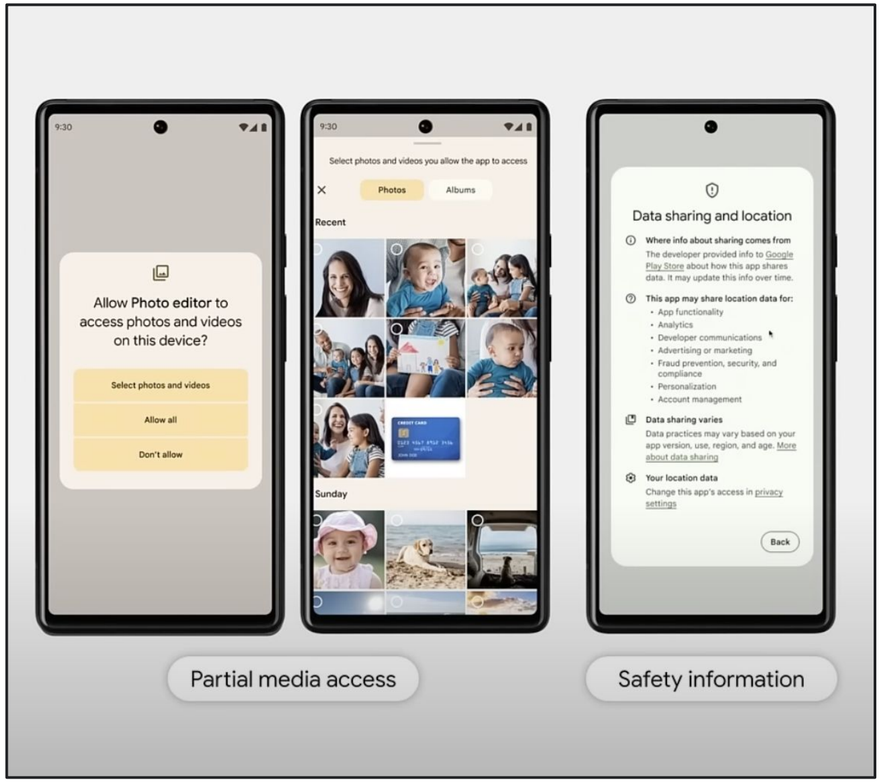
`READ_MEDIA_IMAGES`,

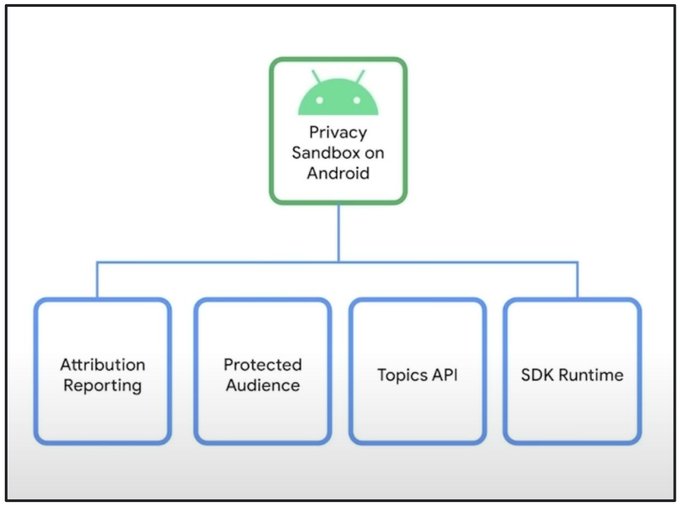
`READ_MEDIA_VIDEO`,

`READ_MEDIA_VISUAL_USER_SELECTED`

Safety information

Informing when apps share users' location data with third-parties





New in Android

Android Studio is more powerful now

Studio bot
Lite mode
Host-side screenshot testing
Compose visual lint
Power profiler
Build Analyzer

and more...

New in Android

Android's modern tooling kit improvements

Latest Jetpack Compose kit got **performance** improvements, **bug fixes** and more **coverage/flexibility** to everywhere you build native UIs.

Moreover text improvements, new layouts and material 3 support.

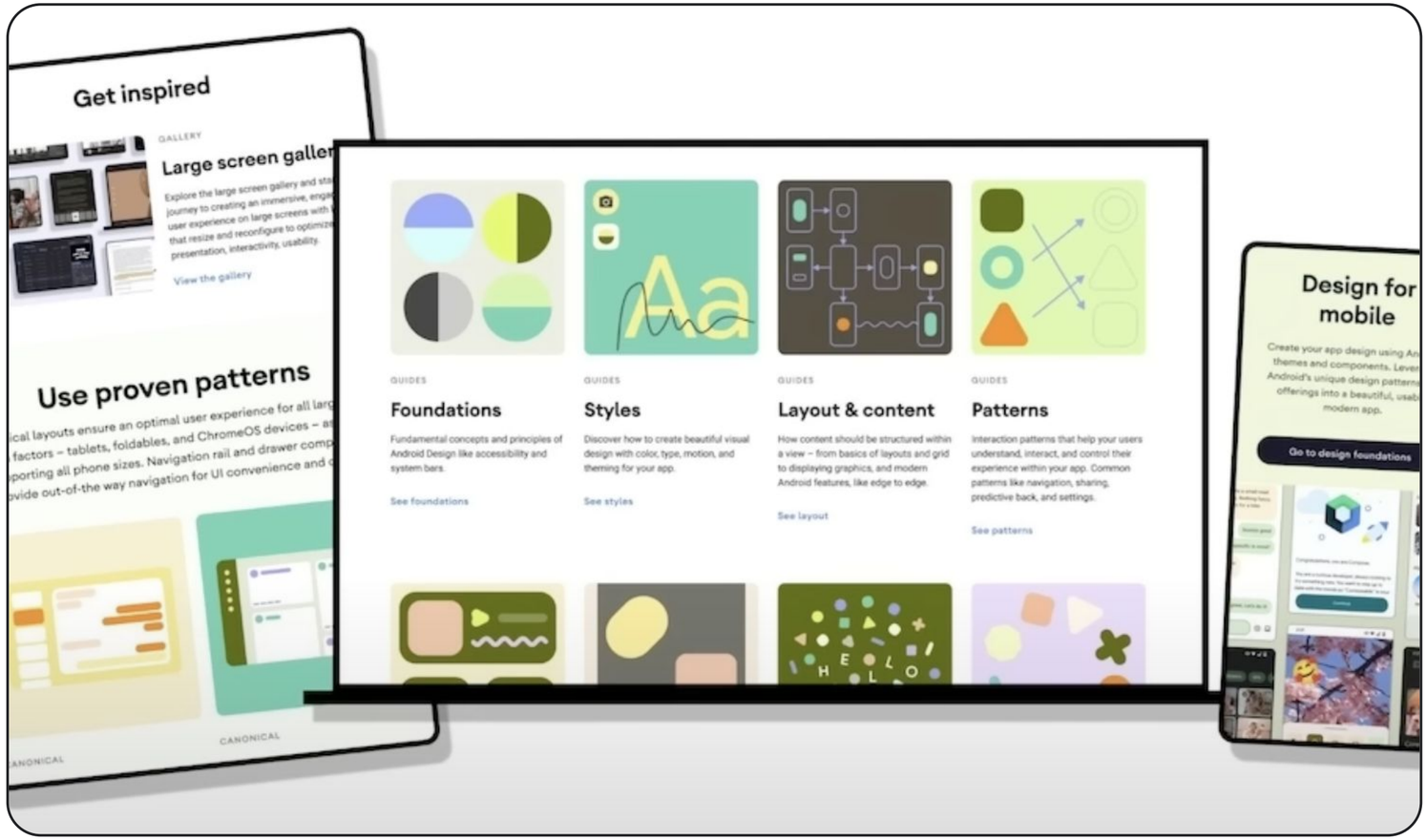
Jetpack Compose

New in Android

New Design Hub

Design beautiful and modern Android apps that meet your user where they are, whether browsing their phone, reading on their tablet, glancing at their wrist, or watching TV.

Visit:
developer.android.com/design/ui



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Android got more...

- Attribution across mobile & web browsers
- ART 14
- Partial Screen Sharing
- Android Auto updates
- Form factors
- Passkeys for passwords
- ML Kit on Android
- Built in Health Connect
- WearOS updates
- Premium camera experiences
- Kotlin compiler performance

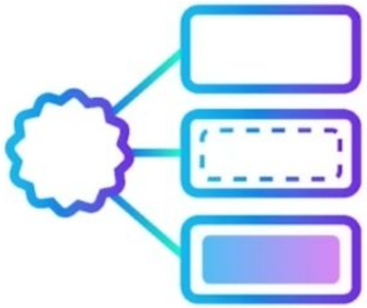
New in Flutter & Dart

Flutter forward continues

Continued focus on developer experience

Breakthrough graphics performance

Seamless integration for web, desktop, and mobile



Early to new and emerging architectures

New in Flutter & Dart

Flutter 3.10 is out

Improved support for Material 3

- Algorithmic colour schemes
- New Drop down
- Navigation drawer
- Tab bar & snack bar updates
- Search bar

Spell check support

Wireless debugging

~~Skia~~ -> Impeller

and more...

New in Flutter & Dart

Dart 3 is out too

100% sound null safety
- Reduced run-time errors
- Improved performance

New language features
- Structured data with pattern matching
- Abstract data types

New class modifiers
- Structured data with pattern matching
- Abstract data types

and more...

Dart 3

```
interface class
base class
final class

sealed class
```

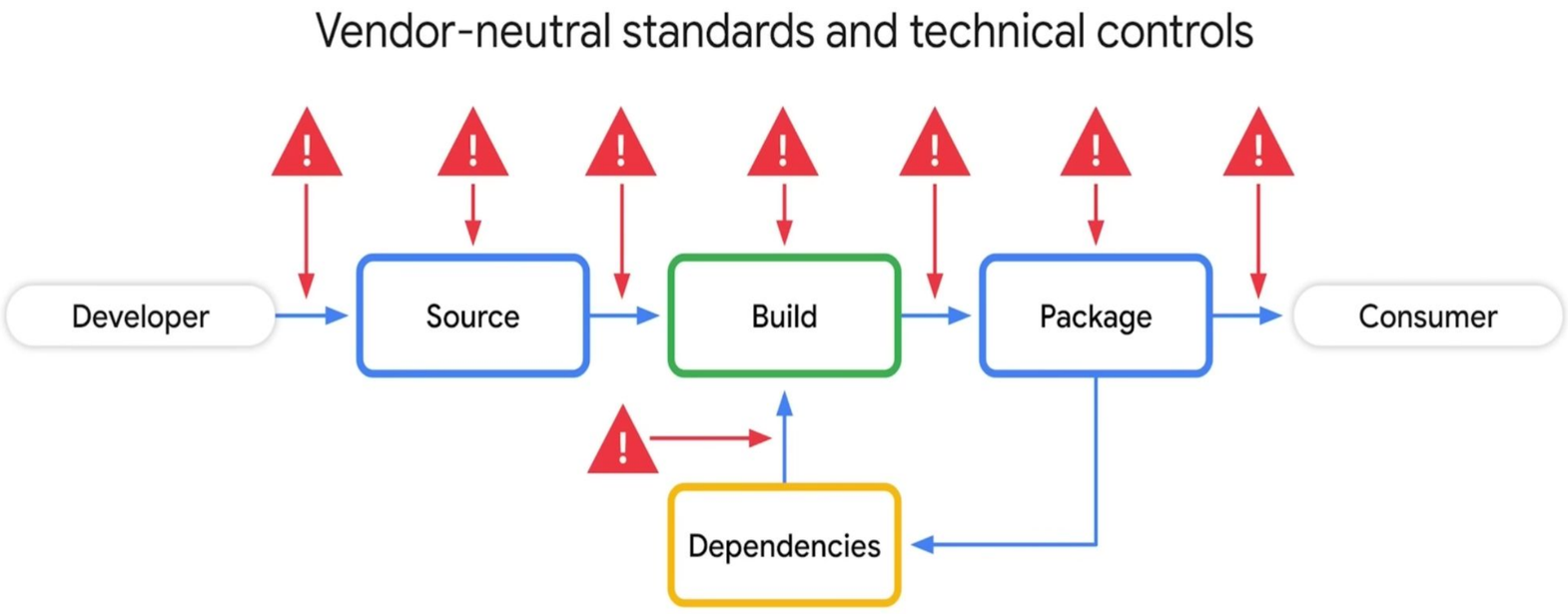
interface class	base class	final class
Disallows extends.	Disallows implements.	Disallows extends and implements.
Useful for defining a contract.	Useful for ensuring all private members are retained.	Useful for closing the type hierarchy.

New in Flutter & Dart

Improved security with SALSA

Supply-chain Levels for Software Artifacts, or SLSA ("salsa").
It's a security framework, a checklist of standards and controls to prevent tampering, improve integrity, and secure packages and infrastructure.

Visit:
salsa.dev



Dart SDK + Flutter SDK on macOS, Linux, and Windows: SLSA L1 ✅

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Flutter got more...

- Breakthrough graphic performance
- Working with structured data with patterns
- Fragment shaders for Flutter web
- Screenshots to pub.dev
- Native ads support
- Improvements to memory page in DevTools
- Topic and tag based search to pub.dev
- Seamless integrations among platforms
- Wide gamut images for iOS

New in Firebase

More power to the Infrastructure

More CF concurrent request handling

Recent improvements to the cloud functions generation 2 can handle upto 1000 concurrent requests and CF V2 is now publicly available for general use.

Cloud function support for Python

Cloud Functions now support python. So you can write your cloud functions in Python and also get the advantage of powerful Python libraries available for mathematical operations and ML.

Cloud firestore improvements

Count has been released for general availability of cloud firestore. QR queries now available for generic querying based on multiple criterias.

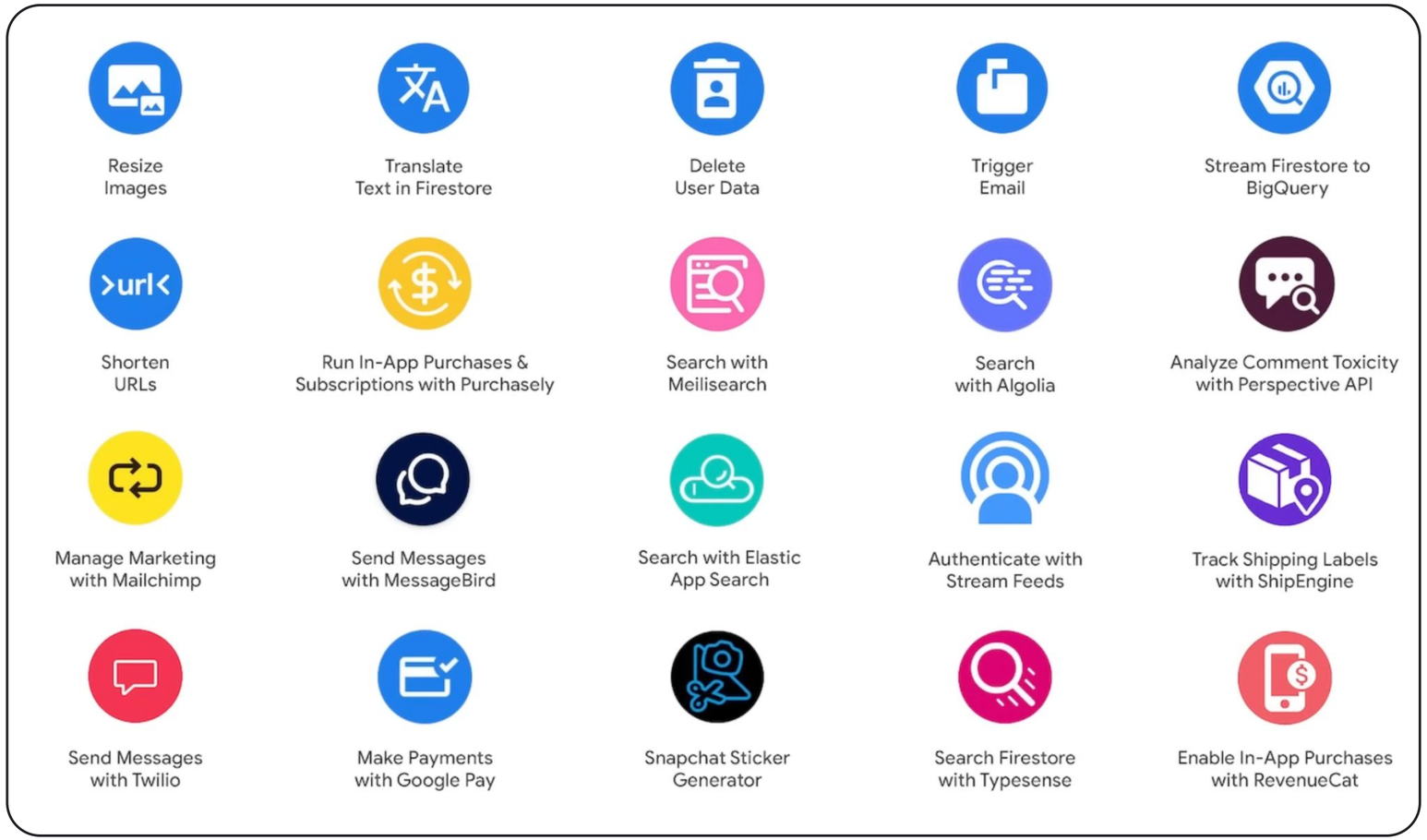
New in Firebase

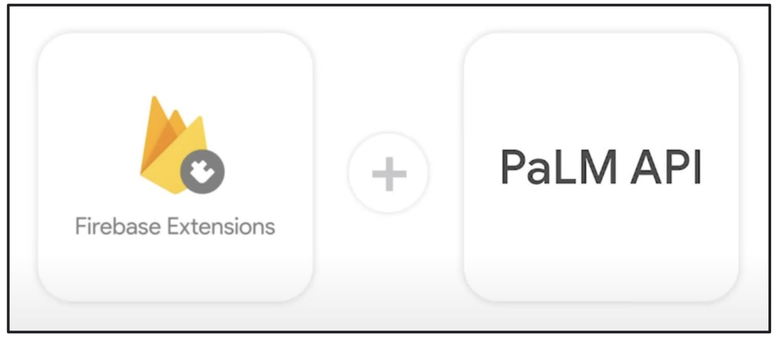
Firebase extensions

Firebase extensions are prepackaged severless solutions that allow you to quickly add new capabilities to your app. Extensions are configurable, updatable and scale automatically.

Moreover, PaLM API extension adds Generative AI capabilities to Firebase. For more extensions,

visit
extensions.dev





New in Firebase

More improvements to Android apps built with Firebase

Deep integrations

Faster Android Startup

A new and efficient thread pool and background workers

Reduced CPU consumption

Deferred product's initialization

~3%

Average performance improvements in time to initial display

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Firebase got more...

- Improvements to Flutter Firebase SDK
- Improved support for more dynamic frameworks
- Firebase integration improvements with Android Studio IDE
- App check improvements
- Firebase support for Terraform
- Improvements to App Distribution
- Firebase hosting support for Flutter web
- App check support for Unity and C++

New in Material

Material 3 APIs are production ready

Production ready **Material U** for compose

Latest jetpack compose support has been released with updated documentation to write beautiful android apps with Materiel U.

Brand new Materiel 3 **components**, **animations** & **themes**

With latest release of m3 we have new components, animations and themes with updated large screen design guidelines for foldables.

visit :
goo.gle/m3-design-kit

Design to code with **Relay**

Design UI components in Figma and use them directly in Compose projects. No more tedious design specs or back and forth to ensure details are right in implemented code.

Visit:
https://relay.material.io

Wallpaper & style
Alarm
06:30
03:30
72°
Timer
Pasta
4:42
09
30
Preview
Come Alive
Google I/O Keynote in 30min
5×5
25
09:30
Alarm will ring at 9:40 AM
12.6k
Edit

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Material got more...

- Monochrome schemes
- Material 3 Design Kit
- Spiritual motions
- Material Figma Integrations
- More support for personalised designs
- Built in prototyping capabilities
- Updated colour fidelity
- Feedback channels to report bugs

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Thank You

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